THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Three identical-looking bars P, Q, R: P attracts Q at both ends, Q attracts R at both ends, R repels P at one end. Which can be concluded? 1

(a) P and R are magnets, Q is magnetic material
(b) all are iron
(c) all are magnets
(d) all are plastic

2. Which of the following is the best argument for using a magnet to separate iron from copper turnings? 1

(a) iron is magnetic and copper is not **(b)** both are shiny
(c) both are metals **(d)** both rust

3. To make a magnet from an iron bar by the single-touch method, the bar should be stroked: 1

(a) in a circle with the magnet
(b) back and forth with both poles
(c) by hitting it with the magnet
(d) always in the same direction with the same pole of a magnet

4. A freely hung bar magnet has one end painted red; the red end settles pointing towards the geographic south. The red end is the: 1

(a) non-magnetic part **(b)** south-seeking pole
(c) north-seeking pole **(d)** neutral centre

5. A mixture contains aluminium, iron and wooden pieces. A magnet separates: 1

(a) iron only **(b)** all metals
(c) iron and aluminium **(d)** wood

6. A student suspends magnet A and brings magnet B's north pole near A's north-seeking end. A turns away. This shows: 1

(a) A is iron **(b)** B has no poles
(c) B's north attracts A's north **(d)** B's north repels A's north

7. Two magnets, each with poles marked, are placed in a line with N of A facing S of B and they attract. If B is reversed, they will: 1

(a) lose their magnetism **(b)** attract as before
(c) repel **(d)** attract more strongly

8. Why does a steel almirah become slightly magnetic over many years? 1

(a) it is painted **(b)** it is exposed to sunlight
(c) Earth's magnetism slowly magnetises it **(d)** it is kept dry

9. **Assertion (A):** A freely suspended bar magnet always comes to rest along the north-south direction. 1
Reason (R): The Earth behaves like a huge magnet.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** An iron nail is attracted by a magnet. 1
Reason (R): Cobalt and nickel are also magnetic materials.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A compass is placed on a steel table and gives a wrong reading. Explain and suggest a remedy. 2

12. A student says that heating a magnet strengthens it. Do you agree? 2

13. Why do we say 'magnets act through non-magnetic materials'? 2

14. Why does an iron nail dangling from a magnet drop other nails when the magnet is pulled away? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how to show that the poles of a magnet are not at the very ends but near them, using iron filings. 3

16. Explain how a loudspeaker uses a magnet. 3

17. How would a magnet help sort iron nails from a mixture of iron nails and brass screws? 3

OR

Explain how the magnetic compass helped in exploring new lands. 3

18. How can you show the Earth's magnetism with a simple experiment? 3

OR

Describe the effect of a steel sheet kept between a magnet and an iron nail. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rohan found an old compass in his grandfather's trunk in Amritsar. The needle always settled pointing in one direction, however he turned the box. When he brought a bar magnet near, the needle moved away from the north. 5
(i) What does the needle of a compass point towards when undisturbed?
(ii) Name the poles of a magnet.
(iii) Why did the needle move when the magnet came near?
(iv) How can a compass help travellers find directions?

20. A box has three unlabelled bars: a magnet, an iron bar and a copper bar. Describe a method with only the bars to identify each. 5

OR

Analyse why magnets lose strength and design a storage plan. 5

END OF QUESTION PAPER — Total: 40 Marks